Osptek Display

TFT LCD SPECIFICATION

Model No:

YDP200B004-V4

1. Introduction

1.1 Scope of application

This specification applies to the LCD module that is supplied by
Shenzhen Osprey Optoelectronics Technology Co., Ltd.
LCD specification: Dots 320xRGBx240
As to basic specification of the driver IC, refer to the IC (ILITEK：ILI9342C) specification and data book.
All material & processing of the LCD module should be Lead Free.

1.2 TFT features:

Structure: TFT PANNEL+IC +FPC+BL;
IPS Type LCD
320 dot-segment and 240 dot-common outputs;
262K Color can be selected by software;
White LED back light;
3SPI+6bit RGB interface

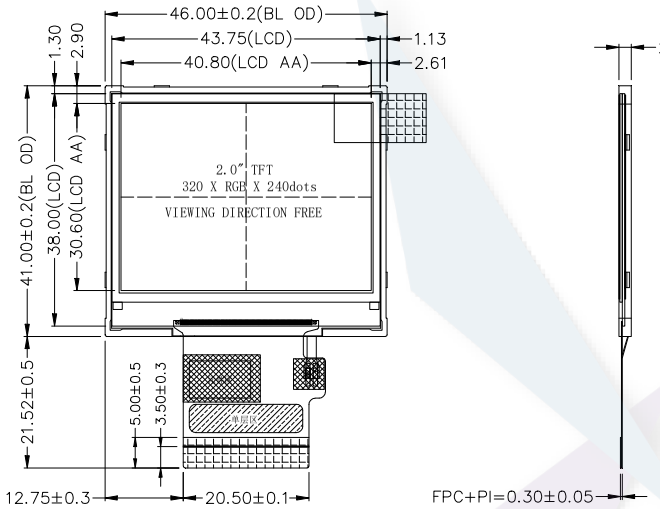
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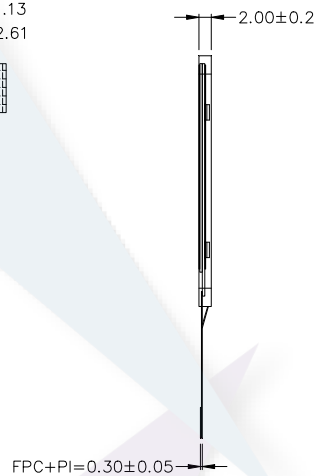
2. LCM General specification

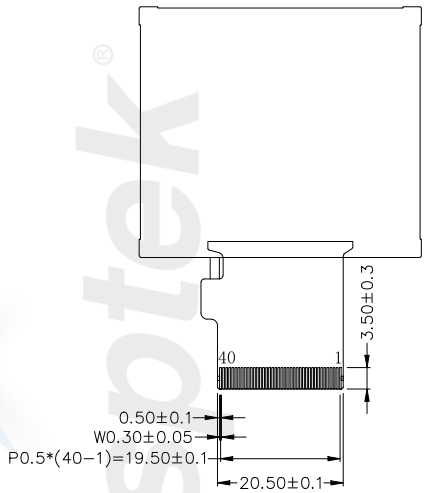
ITEM	Sandard value	Unit
LCD Type	Normally black	--
Drive element	TFT active matrix	--
Number of pixels	320*3RGB(H)X240 (V)	Dots
Pixel arrangement	RGB stripe	--
Pixel Pitch (W*H)	0.1275(H) x0.1275 (V)	mm
Active area	40.80(H) x 30.60(V)	mm
Viewing direction	ALL O'CLOCK	-
TFT Driver IC	ST7789V	-
TFT interface	3SPI+6bit RGB interface	-
Approx. Weight	TBD	g
LCM Size(W*H*T)	46.00(W)x41.00(H)x2.00(T)	mm
Touch structure	--	
Touch Driver IC	--	-
Touch Interface	--	

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REV	DESCRIPTION	DRG DATE	DRAWN BY
	First Desing	2021.05.27	

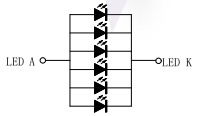






PIN	DESCRIPTION
1	NC
2	NC
3	NC
4	NC
5	NC
6	NC
7	NC
8	NC
9	NC
10	NC
11	NC
12	GND
13	NC
14	VDD
15	NC
16	NC
17	NC
18	NC
19	LEDK
20	NC
21	NC
22	LEDA
23	GND
24	VCI(2.8V)
25	DPI_VSYNC
26	DPI_HSYNC
27	DPI_CLK
28	GND
29	GND
30	DB0
31	DB1
32	DB2
33	DB3
34	DB4
35	DB5
36	SPI_SDI
37	SPI_SCK
38	SPI_CS
39	NC
40	LRSTB

CIRCUIT DIAGRAM



展开出货

NOTES:
1. DISPLAY TYPE: 2.0", 320*240 TFT LCD
2. DISPLAY MODE: transmissive Normally Black
3. VIEWING DIRECTION: ALL
4. DRIVER IC:ILI9342C
5. LCM (White 9 AVG 1/9) :
   Brightness: 600cd/m² (TYP)
   Uniformity:80%(MIN)
6. BACK LIGHT: 6 chip white LEDs If=120mA,Vf=2.9V-3.3V
7. OPERATING TEMP: -20° C TO 60° C,STORAGE TEMP: -30° C TO 70° C
8. * Critical Parameter,( ) ref Parameter,[ ] cpk Parameter
   Unspecified Tolerances:±0.20mm
   Modification mark:
9. SUGGESTION:TP window size unilateral increase 0.3~0.5mm than LCM A.A
10. REQUIREMENTS ENVIRONMENTAL PROTECTION: RoHS

深圳市鱼鹰光电科技有限公司				Mobile:	
PART NO.:	YDP200B004-V4			MODULE NO.:	00
DRAWN BY:	LIU	DATE:	2021.05.27	PROJECTION:	
CHECKED BY:		DATE:		SCALE:N.T.S	UNIT: mm
APPROVED BY:		DATE:		SHEET: 1 OF 1	

Shenzhen Osprey Optoelectronics Technology Co., Ltd. https://www.osptek.com 0755-26810170

3.Absolute Maximum Rating

Characteristics	Symbol	Min.	Max.	Unit
LCM Operating Temperature	T_{OPR}	-20	+60	°C
LCM Storage Temperature	T_{STG}	-30	+70	°C
Humidity	RH	-	90	%

4.Electrical Characteristics

4.1 TFT DC Characteristics

Characteristics	Symbol	Min.	Typ.	Max.	Unit
Supply Voltage for I/O	VDDIO	1.65	1.8	3.3	V
Supply Voltage for(DC/DC)	VDD	2.7	2.8	3.6	V

4.2Back-Light Unit Characeristics

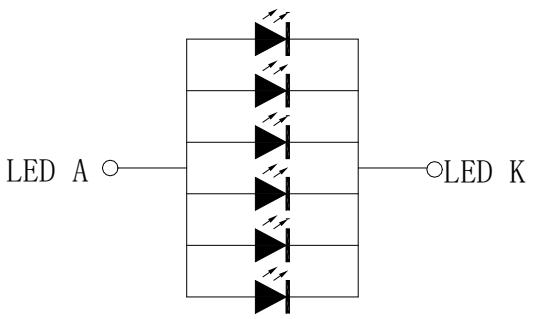
The back-light system is an edge-lighting type with 6 white LEDs. The characteristics of the back-light are shown in the following tables.

Characteristics	Symbol	Min.	Type	Max.	Unit	Notes
Forward Voltage	V_F	2.9	-	3.3	V	--
Forward current	I_F	--	120	-	mA	--
Luminance(With LCD)	Lv	--	600	--	cd/m^2	--
LED life time	N/A	--	30,000	--	Hr	Note 1

Note:

(1) The “LED life time” is defined as the module brightness decrease to 50% of original brightness at I_L=20mA/LED. The LED life time could be decreased if operating I_L is larger than 25mA/LED.

Backlight circuit diagram shown in below:



5. Module Function Description

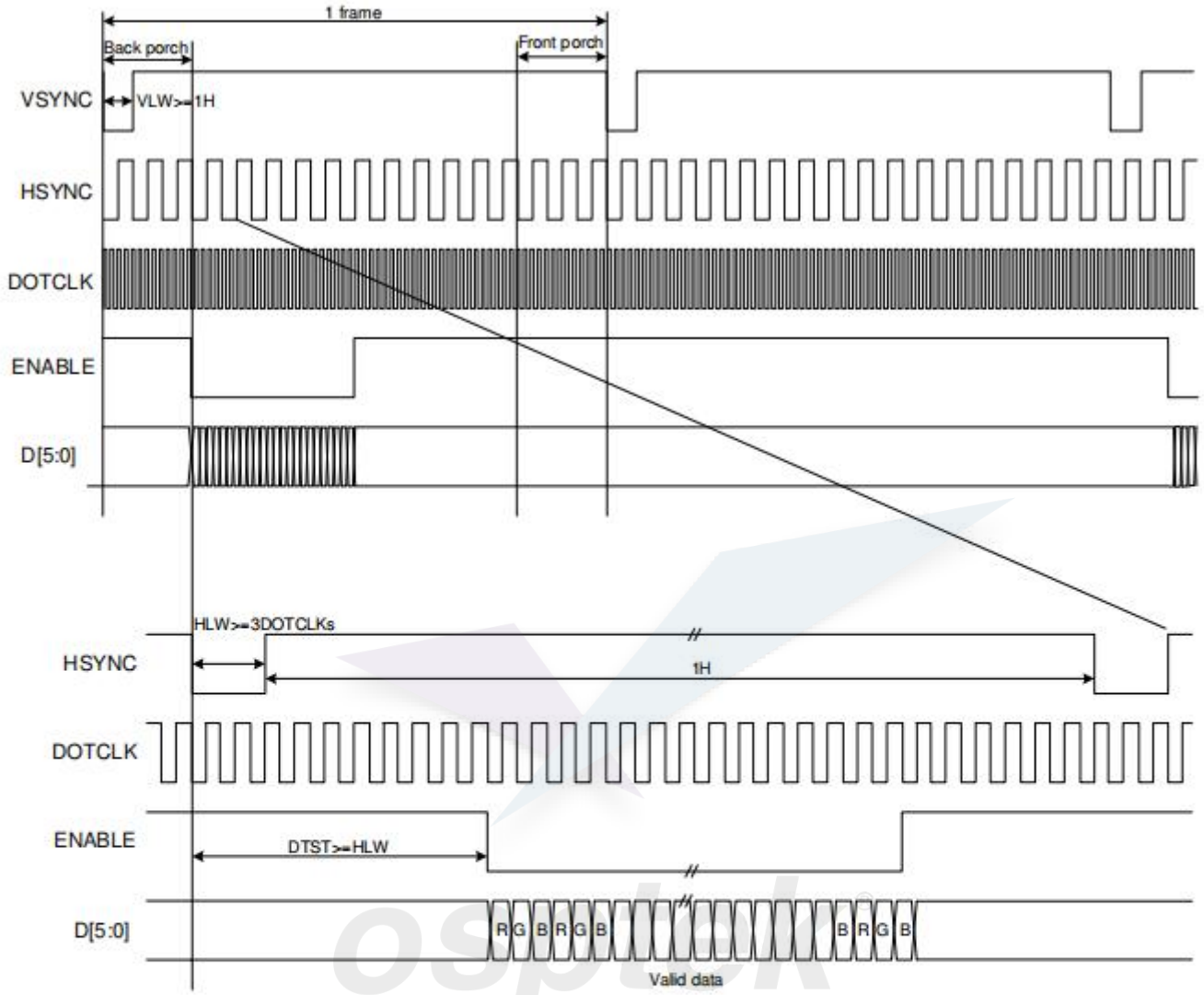
5.1 LCM Pin Descriptions

Pin	Symbol	Description	Notes
1~11	NC	NC	
12	GND	Ground for analog circuit.	
13	NC	NC	
14	IOVCC	POWER SUPPLY I/O (1.8V)	
15~18	NC	NC	
19	LEDK	POWER SUPPLY+ FOR BACKLIGHT CATHODE	
20-21	NC	NC	
22	LEDA	POWER SUPPLY- FOR BACKLIGHT ANODE	
23	GND	Ground for analog circuit.	
24	VCI	PINPOWER SUPPLY (2.8V)	
25	VSYNC	Vertical sync input. Negative polarity.	
26	HSYNC	Horizontal Sync input. Negative polarity.	
27	DCLK	Clock signal. Latching data at the rising edge.	
28~29	GND	Ground for analog circuit.	
30~35	DB0-DB5	6-bit digital data input.	
36	SDA	Serial communication data input.	
37	SCL	Serial commumication clock input.	
38	CS	CHIP SELECT	
39	NC	NC	
40	RESET	This signal will reset the device.	

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6. Timing Characteristics

The timing chart of 6-bit RGB interface mode is shown as below:



VLW : VSYNC Low Width
HLW : HSYNC Low Width
DTST : Data Transfer Startup Time

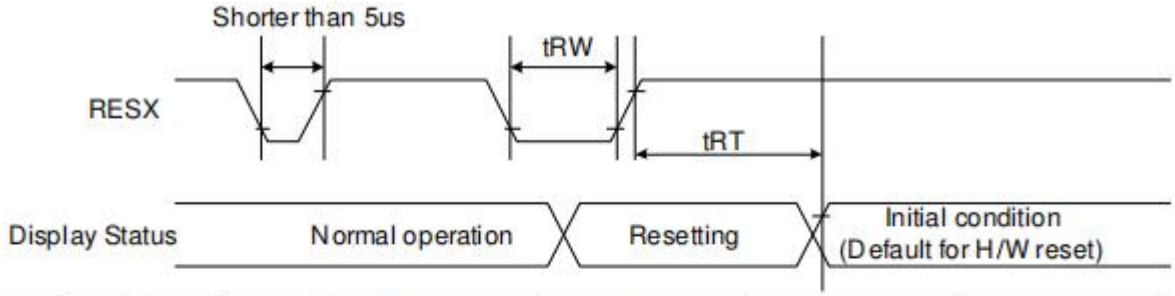
Note 1: The DE signal is not needed when RGB interface SYNC mode is selected.

Note 2: VSPL='0', HSPL='0', DPL='0' and EPL='1' of "Interface Mode Control (B0h)" command.

Note 3: In 6-bit RGB interface mode, each dot of one pixel (R, G and B) is transferred in synchronization with DOTCLK.

Note 4: In 6-bit RGB interface mode, set the cycles of VSYNC, HSYNC and DE to 3 multiples of DOTCLK.

Reset Timing:



Signal	Symbol	Parameter	Min	Max	Unit
RESX	tRW	Reset pulse duration	10		uS
	tRT	Reset cancel		5 (note 1,5)	mS
				120 (note 1,6,7)	mS

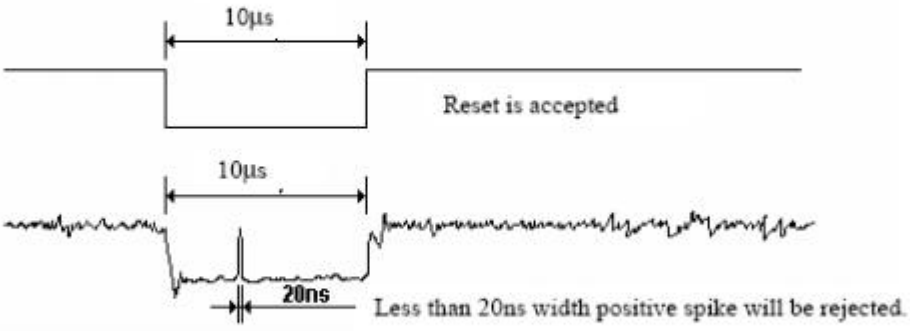
Note 1: The reset cancel includes also required time for loading ID bytes, VCOM setting and other settings from NV memory to registers. This loading is done every time when there is HW reset cancel time (tRT) within 5 ms after a rising edge of RESX.

Note 2: Spike due to an electrostatic discharge on RESX line does not cause irregular system reset according to the table below: -

RESX Pulse	Action
Shorter than 5us	Reset Rejected
Longer than 10us	Reset
Between 5us and 10us	Reset starts

Note 3: During the Resetting period, the display will be blanked (The display is entering blanking sequence, which maximum time is 120 ms, when Reset Starts in Sleep Out –mode. The display remains the blank state in Sleep In -mode.) And then return to Default condition for Hardware Reset.

Note 4: Spike Rejection also applies during a valid reset pulse as shown below:



Note 5: When Reset applied during Sleep In Mode.

Note 6: When Reset applied during Sleep Out Mode.

Note 7: It is necessary to wait 5msec after releasing RESX before sending commands. Also Sleep Out command cannot be sent for 120msec.

7.Optical Characteristics

<Table 4. Optical Specifications >

Parameter		Symbol	Condition	Min.	Typ.	Max.	Unit	Remark
Viewing Angle Range	Horizontal	Θ3	CR>10	70	80	-	Deg.	Note1
		Θ9		70	80	-	Deg.	
	Vertical	Θ12		70	80	-	Deg.	
		Θ6		70	80	-	Deg.	
Contrast ratio		CR	Θ = 0°	1000	1500	-		Note2
Transmittance		Tr		4.34	5.1		%	Note3
Color Gamut		CG		55	60		%	

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit	Remark
White Chromaticity	Wx	Θ = 0°	0.298	0.318	0.338		
	Wy		0.321	0.341	0.361		
Response Time (Rising + Falling)	$T_r + T_f$	Ta= 25° C Θ = 0°		30	40	ms	Note 5

Note:

1.Viewing angle is the angle at which the contrast ratio is greater than 10. The viewing angles are determined for the horizontal or 3, 9 o' clock direction and the vertical or 6, 12 o' clock direction with respect to the optical axis which is normal to the LCD surface (see FIGURE 1).

2.Contrast measurements shall be made at viewing angle of Θ= 0 and at the center of the LCD surface. Luminance shall be measured with all pixels in the view field set first to white, then to the dark (black) state . (see FIGUR 1) Luminance Contrast Ratio (CR) is defined mathematically.

$$CR = \frac{\text{Luminance when displaying a white raster}}{\text{Luminance when displaying a black raster}}$$

3.Transmittance is the Value without APF and without CG.

4.The color chromaticity coordinates specified in the above table shall be calculated from the spectral data measured with all pixels first in red, green, blue and white. Measurements shall be made at the center of the panel.

5.The electro-optical response time measurements shall be made as FIGURE 2 by switching the "data" input signal ON and OFF. The times needed for the luminance to change from 10% to 90% is Tr, and 90% to 10% is Tf.

Figure1 Measurement Set Up

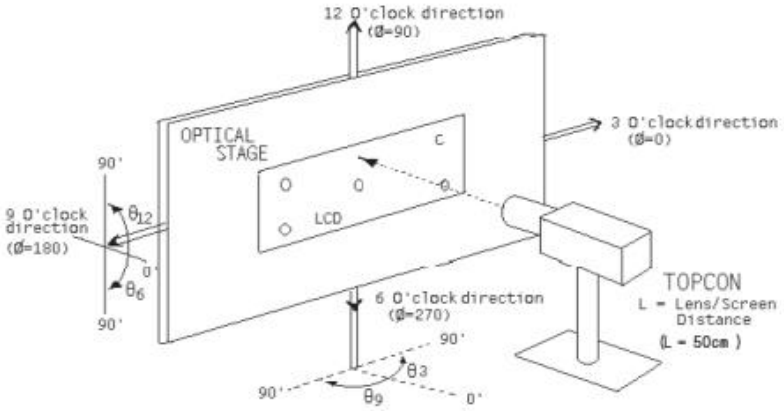
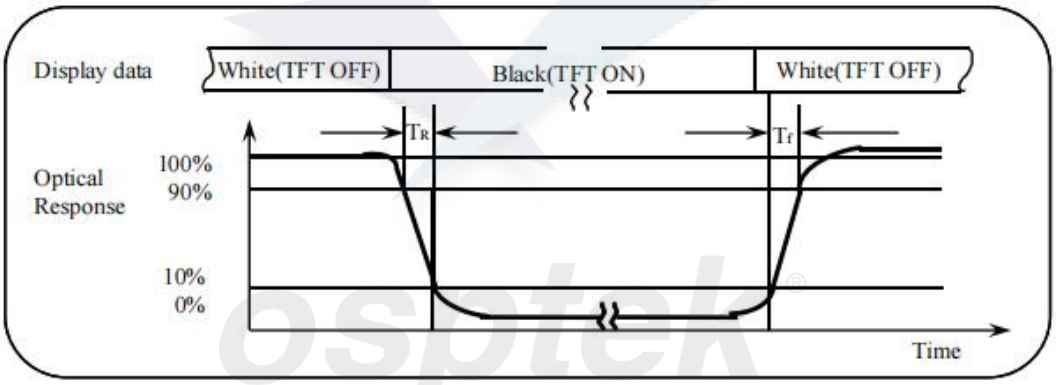


Figure2 Response Time Testing



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8. Reliability Test Item

No.	Test Item	Test Condition	Notes
1	High Temp. Storage	+70°C / 96H	1. Functional test isOK. Missing Segment,short, unclear segment non-display,display abnormally and liquid crystal leakare un-allowed. 2. No low temperature bubbles,end seal loose andfall, frame rainbow.
2	Low Temp. Storage	-30°C / 96H	
3	High Tempe. Operating	+60°C / 96H	
4	Low Tempe. Operating	-20°C / 96H	
5	High Temperature /Humidity storage	50°C x 90%RH /96H	
6	Thermal and cold shock	Static state， -20℃（30min） ~60℃（30min），10 cycles	

9. Packing Method----TBD

- END -

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